Jojy John
Postdoctoral Researcher | Microbiome, Microbial Ecology & Bioinformatics
📧 jojyj@clemson.edu |+1 864-765-5760
🌐 https://jojyjohn28.github.io/ | 💻 GitHub: https://github.com/jojyjohn28| 🔗 LinkedIn: https://www.linkedin.com/in/jojyjohn28

RESEARCH PROFILE

Microbiome researcher with expertise in genome-resolved metagenomics, metatranscriptomics, functional redundancy, viromics, and synthetic microbial community design. My work focuses on understanding how microbial communities maintain functional stability under environmental perturbations and translating these principles toward engineering resilient microbiomes.

ACADEMIC APPOINTMENTS & EDUCATION

Postdoctoral Researcher
Clemson University, USA | 2023–Present
Microbial functional redundancy, metabolic flexibility, estuarine & soil microbiomes, viromes

Ph.D. in Microbiology
Sathyabama Institute of Science and Technology, India | 2021
Genome-resolved studies of halophilic bacteria and stress adaptation

M.S. in Data Science (online) | 2023
Maternity break | Formal training in python, machine learning, and data analysis

M.Sc. & B.Sc. in Microbiology
Mahatma Gandhi University, India | 2014, 2012

TECHNICAL EXPERTISE (Selected)

- **Genomics & Metagenomics:** short-, long-read, and hybrid assembly; MAG reconstruction, binning, and dereplication; genome quality assessment; taxonomic profiling; functional annotation.
- **Metatranscriptomics:** DNA/RNA integration, read mapping, differential expression, and pathway enrichment
- **Viromics:** viral identification, auxiliary metabolic gene (AMG) detection, host prediction, and virus–host interaction networks
- **Metabolism & Modeling:** metabolic reconstruction; functional redundancy modeling; co-occurrence and co-metabolism networks; mixed-effects models; multivariate ecological statistics
- **Computing & Reproducibility:** R, Python, Bash; Snakemake workflows; HPC environments; Git/GitHub; data visualization
- **Machine Learning:** supervised learning for genome binning; random forest models applied to microbial ecology

RESEARCH EXPERIENCE

Postdoctoral Researcher
Clemson University, USA | *2023–Present*

- Led **genome-resolved metagenomic analyses of 105+ samples**, reconstructing **915+ high-quality MAGs** to quantify drivers of microbial community assembly.
- Integrated **metabolic potential with metatranscriptomic activity** (50+ metatranscriptomes; 360 MAGs) to quantify **functional redundancy across 182 metabolic traits** spanning salinity gradients and seasons.
- Designed **synthetic biofilm communities** by assembling and screening **200+ bacterial genomes**, selecting non-pathogenic, non-antagonistic strains with EPS and biofilm traits.
- Developed **automated Snakemake pipelines** for end-to-end metagenomic workflows (QC, assembly, binning, refinement, dereplication) on HPC systems.
- Conducted **viromics analyses** across 36 metagenomes, identifying **1,303 viral genomes** and detecting **auxiliary metabolic genes (AMGs) in ~12.5% of phages**.
- **Mentored graduate and undergraduate** students in multi-omics bioinformatics, including metagenomic/metatranscriptomic workflows, MAG reconstruction, statistical analysis, and manuscript preparation
- Provided **bioinformatics leadership for international collaborations** and managed shared HPC infrastructure (Palmetto2), including software environments and user training.

Project Assistant II (Bioinformatics)
National Institute of Oceanography (NIO), Goa, India | *2019–2021*

- Contributed to an **Indian government–funded project** assessing environmental impacts of deep-sea mining.
- Performed **16S/18S amplicon sequencing and shotgun metagenomics** of deep-sea and hypoxic environments (~600 m; Arabian Sea).
- Participated in **35-day oceanographic research cruises**, including field sampling and onboard DNA/protein preparation.
- Provided **Linux server maintenance and bioinformatics support** and assisted with user training.

Doctoral Research Fellow
Sathyabama Institute of Science and Technology (SIST), India | *2016–2019*

- Conducted **whole-genome sequencing, comparative genomics, and core/pan-genome analyses** of moderate halophilic bacteria.
- Performed **16S rRNA gene sequencing (including Oxford Nanopore)** and phylogenetic analyses.
- Investigated **biodegradation and dye decolorization**, supported by **FTIR-based functional characterization**.
- Cultivated bacterial strains and performed **DNA, RNA, and protein extractions**.

MENTORING

- Supervised or co-mentored **5+ graduate students** (Clemson University, McGill University, SIST, India).
- Mentored **undergraduate and summer research interns** in bioinformatics and viromics.
- Co-author on student-led manuscripts (submitted to **mSphere**, 2026).
- Provided ongoing **lab-wide bioinformatics support** and managed shared HPC/cloud infrastructure (Clemson Palmetto2).
- More details and student reference available: https://jojyjohn28.github.io/mentoring/

PUBLICATIONS

Published

1. **John**, J., Ortiz, M., Ramond, P., & Campbell, B. J. (2026). Functional redundancy and metabolic flexibility of microbial communities in two Mid-Atlantic bays. ISME Communications, ycag021. https://doi.org/10.1093/ismeco/ycag02

2. **John, J.**†, Manigundan, K.†, Radhakrishnan, M., Kumar, A., Abirami, B., & Parli, V. B. (2026). Comparative genomic and phenotypic analysis of Streptomyces rochei SOSIST-3 isolated from the Southern Ocean. Molecular Biology Reports, 53(1), 296. †Co–first authors.

3. Trisha, S., Unnikrishnan, K., Satwik, M., **John, J.,** et al. (2026). Nanoliposome co-delivery of amoxicillin and tazobactam remediates intracellular infection by multidrug-resistant Salmonella enterica serovar Typhimurium. The Journal of Antibiotics-Nature. DOI: 10.1038/s41429-026-00903-5, **Accepted**, waiting for online version

4. **John, J.,** Kumar, A., Alvarez, J., Mannikam, R., Vargas-Chacoff, L., & Leiva Poveda, S. (2026). Draft genome sequence of a novel Winogradskyella species (strain PC D3.3) isolated from a red Antarctic macroalga. Microbiology Resource Announcements. **Accepted,** *proof pending.*

5. **John, J**., Kumar, A., Goddard, M., Manikkam, R., Molina, J., & Leiva, S. (2026). Genomic insights into carbohydrate-active enzymes in a novel Radiobacillus strain from the Antarctic red macroalga *Pyropia endiviifolia.* **Accepted** in World Journal of Microbiology and Biotechnology.

6. Ahmed, M. A., **John, J.,** & Campbell, B. J. (2025). Ecological distribution, environmental roles, and drivers of Actinobacteriota in two Mid-Atlantic estuaries. bioRxiv, 2025.11.

7. Ramadoss, D., **John,** J., Badesab, F., Kocherla, M., Vasanth, A., & Mohandass, C. (2026). Baseline evaluation of the toxicity potential of steel slag using the green mussel Perna viridis for sustainable marine reuse applications. Environmental Geochemistry and Health, 48(2), 90.

8. Ramadoss, D., Ammanabrolu, B. S., Acharya, A., **John, J.,** & Ingole, B. (2025). Deep-sea life associated with sediments and polymetallic nodules from the Central Indian Ocean Basin: Insights from 18S metabarcoding. Deep Sea Research Part II: Topical Studies in Oceanography, 105487.

9. Ramadoss, D., Biju, A., Rathore, C., Saha, M., Kolandhasamy, P., Palogi, C., **John, J.,** & Behera, A. K. (2025). The first report of emerged microplastics in deep-sea sediments: Insights from the Central Indian Ocean Basin. Marine Pollution Bulletin, 211, 117435.

10. **John, J.,** Dineshram, R., Hemalatha, K. R., Dhassiah, M. P., Gopal, D., & Kumar, A. (2020). Biodecolorization of synthetic dyes by a halophilic bacterium Salinivibrio sp. Frontiers in Microbiology, 11, 3281.

11. **John,** J., Siva, V., Kumari, R., Arya, A., & Kumar, A. (2020). Unveiling cultivable and uncultivable halophilic bacteria inhabiting Marakkanam salt pan, India, and their potential for biotechnological applications. Geomicrobiology Journal, 37(10), 1–11.

12. **John, J.,** Siva, V., Richa, K., Arya, A., & Kumar, A. (2019). Life in high salt concentrations with changing environmental conditions: Insights from genomic and phenotypic analysis of Salinivibrio sp. Microorganisms, 7(11), 577.

13. **John, J.,** Siva, V. S., & Kumar, A. (2020). Physiological tolerance of the early life history stages of freshwater prawn (Macrobrachium rosenbergii) to environmental stress. Indian Journal of Geo-Marine Sciences, 49(3), 382–389.

Conference Proceedings

14. **John, J.,** Siva, V., & Kumar, A. (2020). Isolation and characterization of salt pan bacteria for developing bacterial consortia for complete nitrogen removal from aquaculture ponds. In An Insight on Environmental Toxicology (ISBN: 978-93-5396-157-2), pp. 377–388.

Submitted/Ready to submit/Preprints avialable on request

15. Patabandige, D. L. J., **John, J**., Ortiz, M., & Campbell, B. J. (n.d.). Environmental gradients shape the hydrocarbon-degrading microbiome in two Mid-Atlantic bays. **Submitted** (ISME Communications). Preprint available upon request.

16. Chaulagain, D., **John, J.**, Paul, B., Harrington, E. G., McCarthy, G., Sathe, M., Shamabadi, N. S., Carter, E., Leonhardt, J., Nawaz, M. S., Suleman, M., Campbell, B. J., & Karig, D. K. (n.d.). *Genome assemblies of bacterial isolate collections from marine biofilm and water explore microbial diversity*. Genome assemblies (170+) are available in NCBI. **Submitting** to *Microbiology Resource Announcements*

17. Giani, N., **John, J**., & Campbell, B. (n.d.). Metagenomes, metatranscriptomes, and metagenome-assembled genomes (MAGs) collected from soils under different cover crop species. **Submitting** to *Microbiology Resource Announcements*. Metagenomes (n = 21), metatranscriptomes (n = 21), and 355 metagenome-assembled genomes (MAGs) available on NCBI

Selected Conference Presentations (since 2024)

18. Paul, A. M., **John, J.,** Chaulagain, D., Karig, D., & Campbell, B. J. (2025).
Functional redundancy of marine synthetic biofilm communities under different environmental stresses. ASM Biofilms, Oregon, USA.

19. **John, J.,** Ahmed, M. A., & Campbell, B. J. (2025).
Metabolic flexibility, secondary metabolites, and seasonal dynamics in particle-associated and free-living microorganisms in the Chesapeake and Delaware Bays. ASM Microbe, Los Angeles, USA.

20. Campbell, B. J., Kiser, K., Stuckert, S., & **John, J**. (2025).
Bacteriophages and auxiliary metabolic gene interactions in microbial adaptations in the Chesapeake and Delaware Bays. ASM Microbe, USA.

21. **John, J.,** Ortiz, M., & Campbell, B. J. (2025).
Functional redundancy and metabolic flexibility of microbial communities in two Mid-Atlantic bays. ASLO Aquatic Sciences Meeting, Charlotte, USA.

22. **John, J.,** Ortiz, M., Ramond, P., & Campbell, B. J. (2024).
Microbial functional redundancy in response to substrate and energy utilization in estuarine ecosystems. 19th International Symposium on Microbial Ecology (ISME19), Cape Town, South Africa.

TEACHING

- **Co-Instructor**, *MICR 8130: Practical Bioinformatics for Microbiologists* — Clemson University (2026) _Graduate and master's level
- **Instructor**, BIOL 8070: *Readings in Ecology* — Clemson University (2024–2025) _Graduate and master's level
- **Co-Instructor**, MICR 8130: *Practical Bioinformatics for Microbiologists; Hands-on Training in Metagenome Analysis* — Clemson University (2023–2024)_Graduate and master's level
- **Laboratory Assistant**, General Microbiology — SIST, India (2018–2019)
- **Laboratory Assistant**, Microbiology & Biotechnology — Mar Athanasius College, India (2013–2014)

AWARDS

- Best Poster Award, ICWEE (2018)
- Best Oral Presentation, OMICS-AFM Symposium (2018)
- **6th Rank**, M.Sc. Microbiology — MG University (2014)
- **10th Rank**, B.Sc. Microbiology — MG University (2012)

SERVICE & OUTREACH

- Invited talks on microbial ecology and genomics (India)
- Peer reviewer for **20+ manuscripts** (*Microbiology Spectrum, AEM, mSphere, Scientific Reports*, *BMC Microbiology* etc.)

REFRENCES

1. **Barbara J. Campbell, Ph.D.**
   Department of Biological Sciences
   Clemson University
   bcampb7@clemson.edu
   +1 (864) 656-0559

2. **David Karig, Ph.D.**
   Department of Bioengineering
   Clemson University
   dkarig@clemson.edu
   +1 (864) 656-5003

3. **Amit Kumar, Ph.D.**
   Sathyabama Institute of Science and Technology, India
   amitkumar.cccs@sathyabama.ac.in | amit.kumar.szn@gmail.com
   +91-90944-86533